

Looking Ahead: Scaling Open Science

COMMENTARY

Convergence: Detection Tools, Ontologies & Scalability

At a glance: The statistical detection tools described in this guide share a common bottleneck: they operate on unstructured, field-specific, inconsistently reported data. Formal ontologies — machine-readable frameworks that define concepts and their relationships — offer a route to making these tools systematic, interoperable, and scalable. This chapter examines where ontologies are already doing this work, what they require, and where the gaps remain.

1. The Scalability Problem

Each tool described in this guide — GRIM, p-curve, PET-PEESE, bibliometric anomaly detection — was designed by researchers working in a specific domain, solving a specific detection problem with whatever data happened to be available. Each requires the analyst to manually locate the right studies, extract the right statistics, verify that the assumptions are met, and interpret results against field-specific norms. This is scholarly work, not automated screening.

The practical consequence is that despite two decades of methodological development, the detection tools described in this guide remain primarily the instruments of specialists. A meta-scientist running a p-curve analysis on a literature of one hundred studies is doing weeks of work. A journal editor wanting to screen submitted meta-analyses for publication bias has no automated pipeline. A funding agency trying to assess the integrity of a research program has no dashboard. The tools exist; the infrastructure to deploy them at scale does not.

This is not primarily a computational problem — the algorithms are fast once the inputs are in the right format. It is a data representation problem. The inputs are locked in inconsistently formatted PDFs, described with field-specific terminology that varies across journals and decades, and linked to study metadata that is either missing or stored in incompatible systems. Solving the scalability problem means solving the data representation problem first. This is precisely what ontologies are designed to do.

2. What Ontologies Are and Why They Matter Here

An ontology, in the knowledge representation sense, is a formal specification of a domain: a controlled vocabulary of concepts, a taxonomy of how those concepts relate hierarchically, and a set of formal axioms that constrain how instances of those concepts can be combined. The word is borrowed from philosophy (ontology = the study of what exists) but in computer science it refers specifically to machine-readable knowledge structures that support automated reasoning.

Ontologies differ from simple controlled vocabularies (like MeSH or thesauri) in two important ways. First, they express relationships between concepts, not just a list of terms — an ontology can encode that 'randomized controlled trial' is a subtype of 'experimental study', that 'sample size' is a property of a 'study design', and that 'Cohen's d' is a type of 'standardized mean difference effect size'. Second, they support inference: a software agent that knows an ontology can conclude things that are not explicitly stated, by following the logical relationships.

For open science detection tools, three capabilities of ontologies are especially relevant:

- **Interoperability:** if two databases both use the same ontology to describe their studies, a detection tool can query across both without manual harmonization. The FAIR principles (Findable, Accessible, Interoperable, Reusable) depend heavily on ontological annotation to achieve the 'I' — interoperability without human translators.
- **Automated extraction:** an ontologically annotated paper allows software to locate the sample size, the effect size metric, the hypothesis being tested, and the population being studied without reading the paper. This is the prerequisite for automated p-curve construction, bibliometric analysis, and replication matching.
- **Consistency checking:** an ontology-aware system can flag papers where reported values are inconsistent with the stated study design — a form of GRIM-like checking that generalizes beyond integer means to any formally constrained reporting convention.

3. Where Ontologies Are Already in Use

Biomedicine: the OBO Foundry and its ecosystem

The most mature application of ontologies to research practice is in biomedicine, where the Open Biological and Biomedical Ontology (OBO) Foundry coordinates a suite of interoperable ontologies covering organisms, anatomical structures, diseases, phenotypes, chemical entities, and experimental methods. Several of these are directly relevant to research integrity and open science detection:

Ontology for Biomedical Investigations (OBI)	Defines experimental designs, assays, data transformations, and statistical analyses. An OBI-annotated study explicitly encodes its design type, sample size, measurement type, and analysis method — exactly the inputs needed for GRIM, p-curve, and requirements checking.
Human Phenotype Ontology (HPO)	Used in clinical genomics to standardize phenotype descriptions across databases. Enables automated matching of patient cohorts across studies — a prerequisite for systematic replication and cross-study effect size comparison.
Clinical Data Interchange Standards Consortium (CDISC)	Not an ontology per se but a metadata standard widely used in pharmaceutical trials. CDISC-formatted trial data can be queried programmatically for sample sizes, primary endpoints, and outcomes — enabling automated CONSORT compliance checking and outcome switching detection.
Gene Ontology (GO)	The original large-scale biological ontology. Demonstrates at scale what becomes possible when a community commits to a shared vocabulary: cross-database queries, automated annotation, and computational meta-analysis that would be impossible with free-text descriptions.
Research Data Alliance (RDA) vocabularies	Community-maintained metadata vocabularies for data types, repositories, and access conditions. Supports FAIR compliance assessment by providing the controlled terms needed to evaluate whether metadata is machine-readable and interoperable.

Precision medicine: ontologies as infrastructure for evidence synthesis

Precision medicine — the program of matching treatments to patients based on molecular, phenotypic, and contextual profiles — depends on exactly the kind of structured, interoperable evidence base that ontologies enable. The Global Alliance for Genomics and Health (GA4GH) has developed phenopackets, a structured data format for clinical observations that uses HPO, OMIM, and NCIT controlled terms to describe patients in a way that is both human-readable and machine-queryable.

The relevance to open science detection is direct. When clinical trial results are reported using GA4GH phenopackets and CDISC data models, the primary outcomes, patient characteristics, and effect sizes are machine-readable at the point of publication. A system that can read phenopackets can automatically extract the inputs needed to run a p-curve analysis across a therapeutic area, check for outcome switching against a trial registry record, or compute a TES excess significance test — without a human analyst ever reading the full paper text.

The All of Us Research Program and the UK Biobank both use structured phenotype ontologies to annotate their cohorts. This has enabled automated replication studies — computational searches for studies that used comparable populations and endpoints — that would be infeasible with free-text descriptions. The same infrastructure is beginning to be used to flag studies that claim replication of UK Biobank findings but use incompatible phenotype definitions, a subtle form of the outcome-switching problem.

Social and behavioral sciences: emerging standards

The social and behavioral sciences lag substantially behind biomedicine in ontological infrastructure, but several initiatives are building the foundations. The Variable and Value Label Ontology (VVLO) and the Data Documentation Initiative (DDI) provide structured formats for survey instruments and their response scales — directly relevant to GRIM applicability, since the integer scale structure that GRIM requires is machine-readable in a DDI-annotated dataset. The Cognitive Atlas provides a formal ontology of psychological constructs and tasks, enabling automated identification of which studies are estimating comparable effects — a prerequisite for valid p-curve analysis across a literature.

The Psychological Science Accelerator and similar big-team science networks have begun requiring structured metadata at the point of preregistration, using templates that enforce controlled vocabulary for

design type, manipulation type, and outcome measure. When preregistrations use controlled terms rather than free text, automated comparison of preregistered and published outcomes becomes computationally feasible rather than requiring human coding.

4. Core Features of Ontologies Useful for Detection

Not every ontology enables detection tool scalability. Several structural features distinguish those that do from those that merely provide a vocabulary.

Feature	Detection Capability	Example
Feature	What it enables	Example
Formal axioms	Automated consistency checking — flagging impossible combinations of reported values	OBI axioms that constrain which statistical tests are valid for which study designs
Persistent URIs	Stable cross-database linking — the same concept has the same identifier across journals, databases, and decades	OBO PURL system: each term has a permanent URL that never changes
Modular design	Reuse across domains — a study design ontology can import measurement terms from a separate measurement ontology without duplication	OBO Foundry interoperability principle: modules import rather than redefine
Mappings to existing standards	Integration with legacy data — ontology terms are mapped to MeSH, ICD, SNOMED entries so historical literature can be retrospectively annotated	OMOP CDM mappings enable retrospective EHR analysis using ontology queries
Version control	Reproducibility of annotations — specifying which version of an ontology was used at the time of annotation ensures the annotation remains interpretable as terms evolve	OWL format with versioning IRI; BioPortal version history
Community governance	Long-term maintenance and domain validity — concepts are added and revised by domain experts through transparent processes, not unilaterally	OBO Foundry review process; Gene Ontology Consortium

5. The Convergence Opportunity

The detection tools described in this guide and the ontological infrastructure emerging from precision medicine and open science programs are on a collision course — in a productive sense. Each addresses a problem the other cannot solve alone.

The detection tools solve the statistical problem: given a set of study results, how do we diagnose p-hacking, publication bias, analytical fragility, or data fabrication? But they are bottlenecked by the data extraction problem — getting the right numbers from unstructured papers at scale.

The ontologies solve the data representation problem: given a formal description of a study's design, measurements, and results, the relevant inputs for detection tools can be extracted automatically. But ontologies, by themselves, do not perform any detection — they are passive knowledge structures. The detection logic has to be implemented separately.

The convergence opportunity is to connect these two layers: ontologically annotated study metadata feeding automatically into detection tool pipelines, producing systematic, field-wide integrity assessments rather than individual case-by-case analyses. Several specific integration points are already technically feasible:

“The question is no longer whether we have the tools to detect systematic bias at scale. The question is whether the knowledge infrastructure exists to feed those tools without an army of human annotators. Ontologies are the answer to that question — but only if they are adopted upstream, at the point of data collection and reporting, not retrofitted onto the published literature.”

— Synthesized from Wilkinson et al. (2016) FAIR Principles and Ioannidis (2014) on meta-research infrastructure

Integration point 1: Automated p-curve and p-uniform inputs

If published papers annotate their primary outcome tests using a structured reporting ontology (specifying the test type, the test statistic, the degrees of freedom, and the hypothesis direction in machine-readable form), a p-curve or p-uniform analysis over an entire literature can be constructed algorithmically. The Semantic Publishing and Referencing Ontologies (SPAR) — particularly the Statistical Methods Ontology (STATO) — provide exactly this vocabulary. STATO defines t-tests, F-tests, chi-squared tests, and their parameters in OWL, making them queryable across any database that uses STATO annotations.

The Linked Open Data publishing platform for trials, currently piloted by several European research funders, uses STATO annotations in registered and published trials. A proof-of-concept p-curve across all CONSORT-compliant registered trials in a therapeutic area can be computed in minutes once the data is in this format — a task that currently requires months of manual extraction.

Integration point 2: Cross-literature replication matching

One of the most significant open problems in the reproducibility movement is identifying which studies are genuine replications of each other. The Cognitive Atlas and OBI together can support computational replication matching: two studies that share the same Cognitive Atlas task term, the same OBI measurement type, and the same OBI statistical test are strong candidates for replication pairs. This enables automated construction of replication networks — a prerequisite for running z-curve 2.0 or meta-analytic correction methods across a domain rather than a manually curated literature.

The Institute for Replication (i4rep) has begun developing a structured replication registry that uses controlled vocabulary for the original study and replication relationship. As this scales, it becomes possible to automatically compute field-wide replication rates, calibrate TES power estimates against actual replication outcomes, and identify which research programs have systematically low replication rates without requiring a human expert to read each study pair.

Integration point 3: Outcome switching detection at submission

The most promising near-term application of ontology-enabled detection is at the journal submission stage. If a journal requires structured reporting of primary and secondary outcomes in machine-readable form (using CDISC or a compatible vocabulary), and if the trial registry record is also in structured form, the discrepancy between registered and published outcomes can be detected automatically at the moment of submission — before peer review, before publication, and before the paper enters the meta-analytic record.

EQUATOR Network's Registered Report template, combined with the SPIRIT checklist (which has a machine-readable XML implementation), creates the data model for this detection. Several publishers including F1000Research and PLOS Medicine have implemented structured outcome reporting fields in their submission systems. The missing piece is requiring machine-readable registry linkage at submission rather than as a recommendation.

6. Challenges and Open Problems

- Adoption is the fundamental barrier. An ontology that most researchers do not use provides no benefit. Biomedical databases achieved adoption through regulatory requirement (FDA and EMA mandate CDISC for trial submissions) and funder mandate (NIH requires FAIR data management plans). Social and behavioral sciences lack equivalent regulatory leverage, and voluntary adoption has been slow.
- Retrospective annotation is expensive. The existing published literature — millions of papers — is not ontologically annotated and cannot be automatically annotated without error. Large language models (GPT-4, Llama, domain-specific models) are beginning to change this calculus: they can extract

structured metadata from free text at scale, albeit imperfectly. Hybrid workflows combining LLM extraction with ontology-based validation are a promising near-term approach.

- Ontologies age. Concepts evolve, measurement instruments are updated, statistical practices change. An ontology annotation made in 2010 using a term that has since been deprecated is difficult to interpret. The versioning and provenance requirements for useful ontologies are substantially more demanding than for simple metadata schemas.
- Gaming is possible. If journals require structured outcome reporting for detection-tool screening, researchers who wish to misrepresent their analyses could provide structured metadata that does not match the actual analysis. This is a real concern, but it raises the cost and visibility of misrepresentation — structured lies are easier to detect than free-text misrepresentation, because they can be automatically checked for internal consistency and cross-validated against registered protocols.
- Field-specificity limits generalizability. An ontology developed for clinical trials in oncology is not directly applicable to social psychology experiments or economics field experiments. The OBO Foundry's modular approach is the right architectural solution — shared upper-level ontologies with domain-specific extensions — but building those extensions requires sustained community investment in each field.

7. Implications for the Detection Tools in This Guide

Reading the tools in this guide through the lens of ontological infrastructure reveals a gradient of scalability potential.

GRIM / SPRITE / statcheck	Highest near-term scalability. These tools require only summary statistics (mean, SD, N, test statistic). Structured reporting of these values in machine-readable form is already recommended by APA and enforced by some journals. An OBI-annotated paper provides all required inputs automatically.
p-curve / p-uniform	High potential with structured statistical reporting. STATO annotations provide the test statistics and df required. The bottleneck is focal test identification — determining which of many reported tests is the primary outcome — which requires ontological encoding of the study's hypothesis structure, available in OBI but rarely populated.
Funnel plot / PET-PEESE / TES	Moderate potential. Meta-analytic tools require effect sizes and standard errors aggregated across studies. If individual study results use structured effect size reporting (as recommended by the APA Task Force on Statistical Reporting and encoded in STATO), meta-analytic inputs can be assembled automatically from literature search results.
Multiverse / Specification Curve	Lower near-term scalability. These tools require access to raw data and analysis code, not just reported statistics. FAIR data principles and code availability mandates are preconditions, but the additional requirement for structured description of the analytic choice space has no current ontological standard.
Bibliometric Anomaly Detection	Already partially automated. Bibliometric databases (OpenAlex, Semantic Scholar) provide the citation graph and authorship metadata required. The main ontological gap is in author disambiguation (ORCID partially solves this) and topic classification (which determines what counts as an anomalous citation pattern relative to the field).
Outcome switching audit	Highest leverage from ontological infrastructure. The comparison of registered to published outcomes is fundamentally a data matching problem — it requires that both the registry record and the paper use the same controlled vocabulary for outcomes. This is solvable now with existing ontologies; the barrier is policy, not technology.

8. A Path Forward

The convergence of detection tools and ontological infrastructure suggests a practical agenda organized around three time horizons.

Near term (1–3 years): require structured statistical reporting

The highest-leverage immediate intervention is mandating structured, machine-readable statistical reporting at the point of journal submission. This means requiring test statistics, degrees of freedom, effect sizes, and sample sizes in a standardized format — not just as prose in a results section but as structured metadata that can be extracted without natural language processing. Several journals already do this in medical fields; extending the requirement to social and behavioral sciences would immediately enable automated GRIM checking, statcheck screening, and structured p-curve construction across any literature covered by these journals.

Medium term (3–7 years): ontology-enabled replication infrastructure

Building a cross-domain replication registry that uses controlled vocabulary for study designs, populations, manipulations, and outcomes would enable systematic computation of replication rates, z-curve analyses, and TES tests across fields rather than within manually curated literatures. The Institute for Replication, the Psychological Science Accelerator, and equivalents in economics and medical science provide the institutional nuclei. The technical infrastructure — based on existing OBO Foundry and SPAR ontologies — is largely available; what is needed is community agreement on a minimal shared metadata standard for replication registration.

Long term (7+ years): integrated research integrity information systems

The long-term vision is a research information ecosystem in which integrity signals — bibliometric anomalies, p-value distribution warnings, outcome switching flags, GRIM inconsistencies — are computed continuously and attached to papers and research programs as persistent metadata, in the same way that citation counts and altmetrics are currently attached. This is technically feasible given the infrastructure being built in precision medicine and the detection tools now available. The barriers are governance (who manages the integrity metadata?), liability (what are the consequences of a false positive flag?), and incentives (who pays for the annotation infrastructure?). None of these barriers is insurmountable, but none is primarily a technical problem.

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